

Jannet Trabelsi

Phone: (702)807-7207 | Email: JAT332@pitt.edu |
Address: 1700 E Carson St, Pittsburgh, PA, 15203

EDUCATION

University of Pittsburgh, PA

| Expected April 2027

Bachelor of Science in Engineering Physics
Minors in Physics and Electrical Engineering
Certificates in Photonics | Quantum Computing and Quantum Information
GPA 3.912/4.000

Charles University, Prague, Czech Republic

| Summer 2025

Engineering study abroad program

Relevant Coursework

Quantum Computing & Information, Semiconductor Devices, Optics, Electricity & Magnetism, Electronic Circuit Design

RESEARCH INTERESTS

Experimental quantum sensing with solid-state spin systems, specifically NV centers in diamond, for ODMR-based magnetometry, confocal microscopy, and quantum control — with a strong interest in lab automation and scalable instrument control for precision quantum experiments

Manuscripts

First-author manuscript on Adiabatic NV DEER experiment, in preparation

| Expected Fall 2026

RESEARCH EXPERIENCE

Dutt Lab, University of Pittsburgh

| January 2024 – Present

Pulsed Optically Detected Magnetic Resonance Experiment

- Developed a Python-based pulse sequencing and calibration system achieving 0.8 ns timing resolution for multi-channel laser and microwave control, enabling high-fidelity NV-center ODMR measurements
- Simulated the ODMR experiment in software and hardware, with inter-instrument triggering between AWG and counter for data acquisition and optimized experiment for collecting NV centers fluorescence in diamond
- Architected a State Machine ADbasic Program for T1, Rabi, Ramsey, and Hahn echo measurements
- Prepared and wire-bonded nanodiamond and bulk diamond samples for ODMR
- Made IQ calibration to do pulsed ODMR by programming SG384, spectrum analyzer, and Proteus AWG
- Designed and programmed multiharp pulsed experiment for time correlated measurements, gate alignment, hardware jitter/drift detection, and rise time omitting to increase contrast in pulsed experiments.

Continuous Wave Optically Detected Magnetic Resonance Experiment

- Debugged and validated the CW ODMR experiment, identifying and fixing critical software errors that were preventing operation; collected ODMR data on nanodiamond and bulk diamond samples, with and without an applied magnetic field
- Achieved CW ODMR with Single NV centers in diamond

Confocal Microscopy and Positioning Experiment

- Built a Python suite to automate and optimize NV-center count collection in confocal microscopy
- Designed a modular Python GUI for sample positioning and data analysis, enabling reproducible confocal scans of diamond NV centers
- Developed an automated EMCCD camera scan with microdrive and autotune, and implemented wide-field optics and white-light imaging that acquires the full field in a single exposure versus sequential point-by-point scanning over ~810,000 positions
- Performed confocal alignment and optimization for NV-center imaging; cleaved samples and prepared nanodiamond solutions; collected photoluminescence data via spectrometer

Lab Software Project

- Programmed servers with FastAPI, allowing all lab computers to access any hardware across the lab
- Developed Python-based instrument control suite (FastAPI servers, GPIB universal frequency counter, USB camera integration via DLL, spectrum analyzer GUI with FFT visualization, and micro-stage automation), reducing acquisition time by 50%, enabling unified data collection, data saving, and repurposing lab hardware
- Networked laboratory devices, making all devices accessible across all lab computers
- Developed and tested Data Analysis GUI for data visualization and data plotting, compatible with lab wide hdf5 data saving, confocal microscopy, sample positioning, Rabi, Ramsey, Pulsed ODMR, CW ODMR, and Hahn Echo experiments data

Spin Charge Conversion Experiment

- Aligned and tuned a dye ring laser cavity
- Programmed spectrometer, wavemeter, and dye laser wavelength tuning
- Designed and aligned optical setups for spin charge conversion experiment
- Repaired and refurbished Dye Circulator and Water Chiller Circulator

Annealing Chamber Project

- Designed and prototyped an annealing chamber, performed computational fluid dynamics and thermal simulations using SOLIDWORKS, and communicated instructions with machinists

Levitating Diamond Experiment

- Designed a vacuum re-entrant flange and performed static simulations for Ultra High Vacuum window using SOLIDWORKS to use a higher Numerical Aperture microscope objective, improving imaging of levitated particles by 43%
- Developed an Arduino-based PID control system laser thermal stabilization in a levitating diamond setup

General Lab

- Aligned and fiber-coupled lasers for spectrometer calibration
- Authored SOPs and training guides, accelerating onboarding of incoming researchers and enhancing lab safety
- Presented progress in weekly group meetings

HONORS

TRIO 2024 National Scholarship Award

| October 2024

Gloria and Mark Tressler Scholarship for the Swanson School of Engineering

| October 2025 – April 2027

Swanson School of Engineering Honor Student

| August 2024 – Present

Dean's List

| August 2023 – Present

Sigma Pi Sigma Physics Honor Society

| Spring 2026 – Present

TECHNICAL EXPERTISE

Instrumentation & Control

- Python (Instrument control, automation, GUIs), ADbasic, FastAPI servers, GPIB/USB/LAN/RS232/PXI instrument control, PID control, waveform generation

Hardware & Electronics

- Altium PCB design, LTspice digital/analog circuit design, VHDL, Quartus, Soldering

Simulation & CAD

- SOLIDWORKS (CFD, Static, Thermal), ANSYS Mechanical, Granta EduPack, Autodesk

Quantum & Optics Experiments

- Confocal Microscopy, NV Centers, AOMs, Wavemeters, Spectrometers, Cryostats, Vacuum Systems, PID Control, Interferometry, Annealing Chamber, Arduino, Nanodrive, Actuators, TEC, Universal Counters, Spectrum Analyzer, EMCCD Camera, Arbitrary Waveform Generators

Programming

- Python, C++, MATLAB, ADbasic, JSON, HTML, Git

CLASS PROJECTS

Quantum Directed Research, University of Pittsburgh

| January 2026 – Present

- Applied group theory to analyze defect symmetries in SiC and NV0/NV– centers (directed study with Dr. Devaty)
Optics, University of Pittsburgh | November 2025
 - Executed optical pumping experiment with a Nd:YAG crystal, including alignment, data acquisition, and analysis
 - Prepared Michelson interferometer and measured coherence length and coherence time for various light sources.
 - Investigated single-photon interference phenomena through experimental measurement using a photomultiplier tube and quantitative analysis
- ECE Junior Design** | January 2026 – April 2026
- Designed, prototyped, built, and tested three Printed Circuit Board projects: educational dummy spectrometer, Quantum Bop It to teach Quantum Computing in a team environment, and hybrid analog-digital Acousto-Optic Modulator RF Driver to control laser

LEADERSHIP AND COMMUNITY ENGAGEMENT

- Volunteer, Quantum Computing Outreach Event** | September 2024 – September 2025
- Organized a quantum mechanics and computing outreach event, engaging 19 middle school students through interactive sessions
 - Presented dye laser system
 - Led activities including quantum computing games to inspire interest in the quantum field
- Member, Panther Racing Pitt FSAE** | Fall 2024 – Spring 2025
- Analyzed low-voltage battery telemetry and optimized wiring harness design, reducing vehicle mass by 2.5 lb

PROFESSIONAL EXPERIENCE

- Philanthropic and Alumni Engagement, University of Pittsburgh** | September 2023 – September 2025
- Administrative Support**
- Coordinated logistics for alumni donations and engagement events
 - Developed communication and interpersonal skills to ensure that donation processes are efficient and secure

LANGUAGES

Arabic: Native
 English: Professional / Fluent
 French: Professional / Fluent